Music Genre Classification: A Dilated CNN as Feature Extractor for Classical Classifiers

Reproduction and extension of Li, Xue & Zhang, *Combining CNN and Classical Algorithms for Music Genre Classification* (Stanford CS229)
ML mini-project

1. Problem statement

Given a 30-second music clip, predict its genre among five classes: blues, classical, hip-hop, metal and pop. The reference paper trains a small *dilated CNN* on MFCC features and then feeds the CNN's intermediate-layer activations to classical classifiers (Logistic Regression, GDA, Random Forest, SVM). It claims this greatly improves the classical models and can even beat the CNN itself. Our goal was to **reproduce** the paper's results, test how **reliable** its claims are, and **improve** on it where possible.

2. Dataset

We use **GTZAN** (Tzanetakis & Cook), restricted to the paper's five genres: 5 genres × 100 clips of 30 s = **500 clips**. Each clip is converted with *librosa* into a 20 × 1290 MFCC matrix (25,800 features when flattened). For the reproduction we use a fixed stratified **400/100 train/test split** (seed 42). Because 100 test clips means one clip is worth 1% accuracy, every later comparison uses **stratified 5-fold cross-validation** over all 500 clips (mean ± std).

3. Approach

- **Baselines:** LR, GDA (LDA with shared covariance), Random Forest (depth 7) and RBF-SVM on raw flattened MFCCs and on PCA-50.
- **Dilated CNN:** two units of [dilated Conv1d, ReLU, AvgPool, Dropout, BatchNorm] followed by a 5-way softmax layer. Unit 1: 8 filters, kernel 8, dilation 4, pool 32 (**320** features); unit 2: 24 filters, kernel 4, dilation 2, pool 16 (**48** features). Adam (lr 1e-3), cross-entropy, up to 100 epochs.
- **CNN as feature extractor:** flatten the layer-1 and layer-2 activations and train the four classical models on them.
- **Extensions:** (i) leakage-free 5-fold CV; (ii) GDA with covariance shrinkage; (iii) dilation ablation (dilation 1 vs 4/2); (iv) hyperparameter tuning by inner cross-validation; (v) **3-second segments**: each clip is cut into ten 129-frame segments, the CNN is trained on segments, and segment predictions are combined per clip (mean softmax for the CNN, majority vote for classical models).

4. Implementation overview

Python with PyTorch, scikit-learn and librosa, run on Kaggle (CPU). The repository contains *data.py* (MFCC extraction, caching, split), *baselines.py*, *dcnn.py* (model and training), *extract_features.py*, *cv.py* (cross-validation and dilation ablation), *tuning.py* (GDA shrinkage and tuning), *segments.py* and *predict.py* (demo: audio file in, genre and class probabilities out, using saved weights). To avoid leakage, scalers, PCA and MFCC normalisation are fitted on training data only; the best CNN epoch is chosen on a validation split taken from the training data (the paper tuned on the test set); the CNN is retrained inside every CV fold; and segment folds are created per *clip* before cutting, so no song appears on both sides of a split. Cross-validation and the dilation ablation were run by a teammate; the remaining code and experiments are the author's.

5. Results (test accuracy)

Fixed split: ours (paper)	LR	GDA	RF	SVM
Raw MFCC (25,800)	0.70 (0.77)	0.81 (0.77)	0.79 (0.76)	0.77 (0.28)
CNN layer 1 (320)	0.80 (0.87)	0.65 (0.61)	0.87 (0.86)	0.86 (0.87)
CNN layer 2 (48)	0.86 (0.84)	0.89 (0.80)	0.86 (0.87)	0.89 (0.84)

Table 1. *Reproduction on the 400/100 split. The CNN alone scores* ***0.84*** *(paper: 0.84-0.87). The paper's SVM collapse (0.28) appears only with gamma='auto' on unscaled features (ours: 0.32); with the current default the same SVM reaches 0.77.*

5-fold CV (mean ± std)	LR	GDA	RF	SVM
Raw MFCC (25,800)	0.814 ± 0.032	0.800 ± 0.043	0.790 ± 0.037	0.824 ± 0.037
CNN layer 1 (320)	0.846 ± 0.024	0.582 ± 0.036	0.842 ± 0.031	0.856 ± 0.030
CNN layer 2 (48)	0.868 ± 0.029	0.840 ± 0.054	0.872 ± 0.036	0.866 ± 0.025

Table 2. *Cross-validated accuracy (500 clips). The full-clip CNN alone scores* ***0.846 ± 0.050****.*

Extension	Result (5-fold CV)
GDA shrinkage (layer 1)	GDA 0.582 to **0.808** ± 0.026, better in 5/5 folds; no clear gain on layer 2 (0.840 to 0.850)
Dilation ablation	CNN alone 0.846 ± 0.050 (dilated) vs 0.828 ± 0.035 (none); layer-2 classifiers differ by at most 1.2 points: **no clear effect**
Hyperparameter tuning	LR / RF / SVM change by -1.0 to +1.2 points, inconsistent across folds: **no reliable gain**
3 s segments + voting	CNN alone **0.876 ± 0.021** (per segment 0.826); LR on layer 1: **0.886 ± 0.029**; RF on layer 2: **0.884 ± 0.027**

Table 3. *Effect of each extension. Segment-model layers have 128 and 96 features (smaller pooling), so its gain mixes more data with a changed pooling size.*

Error analysis. On the fixed split, metal and pop are always recognised and classical almost always (19/20); blues and hip-hop clips are most often mistaken for metal (5 clips each).

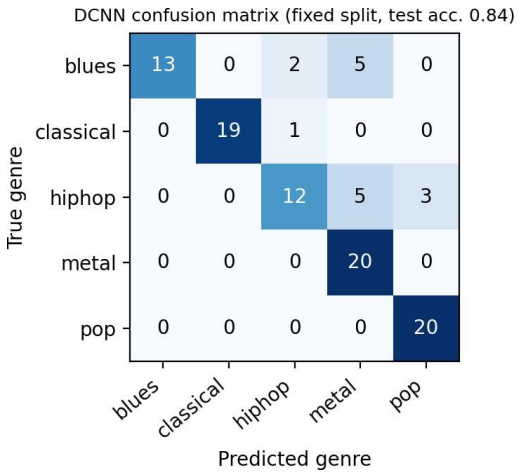


Figure 1. *Confusion matrix of the full-clip dilated CNN on the 100 held-out clips (rows: true genre, columns: predicted).*

6. Conclusions

- **The paper's main idea holds:** classical models on the CNN's layer-2 features reach about 0.87 versus 0.79-0.82 on raw MFCCs, a gain of roughly 4-8 points in cross-validation (larger on the fixed split).
- **Parts of the paper's story are weaker than reported:** its SVM baseline is an artifact of a hyperparameter setting, and classical models beat the CNN itself by only about 2 points, within fold-to-fold noise (std 3-5 points).
- **Improvements:** GDA shrinkage repairs GDA on high-dimensional features, and training on 3 s segments with per-clip voting raises accuracy to about 0.88 and halves the CNN's variance across folds (std 0.050 to 0.021), though much of its average gain comes from one weak fold (0.76 to 0.86).
- **No clear benefit** was found from dilation or from hyperparameter tuning at this data size.
- **Limitations:** one training seed per fold; GTZAN is small with known quirks; the dilation test did not control for the receptive-field span; unit-2 settings are our choice to match the paper's feature counts.

References. [1] Li, Xue, Zhang, *Combining CNN and Classical Algorithms for Music Genre Classification*, Stanford CS229. [2] Tzanetakis & Cook, Musical genre classification of audio signals, IEEE TSAP 10(5), 2002.